

Muhammad Afsah Mumtaz

AI Engineer | Quantitative & Fintech Systems | Multi-Agent Trading

Dubai, United Arab Emirates | Open to Dubai / Abu Dhabi (on-site or hybrid) and remote worldwide

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Nationality: Pakistani | Visa: UAE Golden Visa (no sponsorship required) | Notice period: Immediate

PROFESSIONAL SUMMARY

AI engineer building **quantitative, agentic trading systems** (BSc Hons Information Technology student, Middlesex University Dubai). Designed and shipped **Pantheon-Trades** — a live 11-agent large language model council that debates every prediction-market trade on Polymarket with **Half-Kelly position sizing, self-calibrating agent weights (Platt / isotonic regression), an on-chain audit trail, and a 200-market backtest (Brier 0.149)**. Also fine-tuned a 2B LLM to 82% ARC-Challenge on a single 4 GB GPU. Strong in Python, FastAPI, Solidity, PyTorch, and risk-aware system design. Seeking an **AI / quantitative / fintech engineering internship or junior role**. On-site or hybrid in Dubai / Abu Dhabi or remote worldwide; UAE Golden Visa holder — no sponsorship required, immediate start.

TECHNICAL SKILLS

Quantitative & Trading: Algorithmic Trading, Backtesting, Strategy Evaluation, Position Sizing (Kelly Criterion), Risk Management, Probability Calibration (Platt / Isotonic), Brier Score, Prediction Markets, Portfolio / Risk Constraints, Avoiding Look-Ahead Bias

AI / Machine Learning: Large Language Models (LLMs), Multi-Agent Systems / Agentic AI, LLM Orchestration, Retrieval-Augmented Generation (RAG), Fine-Tuning (LoRA / QLoRA), Model Evaluation, Prompt Engineering, PyTorch, Hugging Face Transformers, Ollama

Blockchain & Web3: Solidity, Foundry, Smart Contracts, On-Chain Settlement, Arc Testnet, Polymarket CLOB

Languages: Python, Rust, TypeScript, JavaScript, C++, SQL

Backend & Data: FastAPI, Node.js, Express, REST APIs, PostgreSQL, Redis, MongoDB, SQLite

Infra & Tooling: Docker, Git, GitHub Actions, CI/CD, Turborepo / pnpm, Testing (pytest, Foundry)

PROJECTS

Pantheon-Trades — Multi-Agent Quant Trading Platform (Live: pantheon-trades.vercel.app)

Tech: Python, FastAPI, Solidity / Foundry, PostgreSQL, Redis, Next.js 14, Multi-LLM (Claude, Gemini, OpenAI, +8)

- Designed and shipped a live platform where an 11-agent LLM council debates every binary prediction-market trade on Polymarket across four structured rounds before execution.
- Implemented risk-aware execution: Half-Kelly position sizing, constitutional position caps (5% NAV max), unilateral risk vetoes, and an on-chain proof-of-restraint audit trail recording rejected trades.
- Built self-calibrating agent weights from Brier scores with council probabilities recalibrated via Platt / isotonic regression; validated on a 200-market backtest (council Brier 0.149).
- Hardened with 714 Python and 65 Solidity tests across a pnpm / Turborepo monorepo, demonstrating production-grade reliability and testing discipline.

AVA — LLM Fine-Tuning, Evaluation & Inference Lab

Tech: Python, PyTorch, Hugging Face Transformers, PEFT / QLoRA, RAG, Ollama, GGUF

- Fine-tuned Qwen3.5-2B with a 42 MB QLoRA adapter to 82.0% on ARC-Challenge and 92.0% on ARC-Easy, outperforming Llama 3.2 3B-Instruct (78.6%) — all on a single 4 GB GPU.
- Built a reproducible evaluation harness across 17 benchmarks (16,872 tasks) with statistically careful scoring; training peaked at 1.81 GB VRAM in 100 minutes.

MALD — Local AI Knowledge System with Cited RAG

Tech: Rust, SQLite FTS5, HNSW vector indexing, RAG, Ollama

- Built a privacy-first knowledge system as a single Rust binary with hybrid full-text and semantic (HNSW) search and cited RAG chat referencing exact files and lines.

pane — Windows-Native Linux Environment Platform

Tech: Rust, WSL2, XRDP, PowerShell, Windows packaging automation

- Developed a Windows-native control surface to create, launch, repair, and manage Linux desktop environments — shipped an Arch-first MVP packaged as pane.exe with automated release packaging.

Entangled — Full-Stack Web Application with AI Features

Tech: JavaScript, Node.js, Express, MongoDB, REST APIs

- Built a full-stack single-page app (auth, feeds, messaging, search, uploads) over a REST API backend, with integrated AI chatbot, summarization, and code-explanation features.

EXPERIENCE

Open-Source Contributor, HAGI (Hypercomplex AGI Research) 2026 — Present
github.com/ShmidtS/HAGI | Remote

- Contribute on the experimental branch of an open research architecture exploring Clifford-algebra reasoning in small language models, working across PyTorch, Rust, and Lean4 in a collaborative codebase.

Chairman, MDX Computing Society Sep 2025 — Present
Middlesex University Dubai | Dubai, UAE

- Lead the university computing society across technical initiatives, events, and community direction.

Committee Chair, BCS Student Chapter Sep 2025 — Present
BCS, The Chartered Institute for IT | Dubai, UAE

- Chair the BCS student chapter committee, planning and executing student-led technical and professional initiatives.

Student Voice Leader Oct 2024 — Present
Middlesex University Dubai | Dubai, UAE

- Liaise between students and the programme team on teaching quality, resources, assessments, and student experience, representing consolidated cohort feedback in academic forums.

Customer Service Representative Oct 2021 — Jan 2022
transcosmos inc. | Debrecen, Hungary

- Resolved customer issues in a structured service environment requiring communication discipline and resilience under pressure.

EDUCATION

BSc (Hons) Information Technology Sep 2024 — Present
Middlesex University Dubai | Dubai, UAE

A Levels Completed
Cambridge International School, Dubai | Dubai, UAE

CERTIFICATIONS

- IBM AI Engineering Professional Certificate — Coursera
- Google IT Automation with Python Professional Certificate — Coursera
- Foundations of Project Management — Google / Coursera
- Data Visualization and Dashboards with Excel and Cognos — IBM / Coursera

ADDITIONAL INFORMATION

Languages: English (Native), Urdu (Native), Arabic (Professional Working), Hungarian (Limited Working)
Work Authorization: UAE Golden Visa holder — no sponsorship required
Availability: Internship, junior, contract, freelance, and full-time roles. On-site or hybrid in Dubai / Abu Dhabi, UAE, and remote worldwide. Immediate start.